

EDUCATION

TECHNICAL UNIVERSITY OF MUNICH <i>Robotics, Cognition, Intelligence (M.Sc.)</i>	Munich Apr 2026 – Present
TECHNICAL UNIVERSITY OF MUNICH <i>Bachelor of Informatics with Minor in Economics</i> <i>Bachelor's Thesis: "Development of a 3D Human-Machine Interface for Construction-Site Robotic Task Planning"</i> • Browser-based 3D interface with real-time digital-twin visualisation, robot and task configuration, trajectory preview and collision checking (React, Three.js, TypeScript, ROS 2 Jazzy) • Integrated task planning and safety sign-off in the browser; collaborated with Sitegeist Robotics and the Professorship of Cyber Physical Systems, TUM	Munich Oct 2022 – Feb 2026
KARLSRUHE INSTITUTE OF TECHNOLOGY, STUDIENKOLLEG <i>T-Course + Informatics – Grade: 1.7 (German grading system)</i> • Deutsche Physikalische Gesellschaft Abiturpreis Award	Karlsruhe Feb 2021 – Feb 2022

ROBOTICS & AI EXPERIENCE

CAPGEMINI <i>Working Student Robotics Engineer</i> <i>NVIDIA Omniverse (Isaac Sim, Isaac Lab), Python, Docker (Compose)</i> • Develop control software for the Unitree G1 humanoid robot, from simulation to deployment on hardware • Own an end-to-end data pipeline: collection via simulation and teleoperation, conversion, fine-tuning, deployment	Munich Jan 2026 – Present
PRACTICAL COURSE: ROBOTIC ARM DESIGN OPTIMIZATION <i>Professorship of Cyber Physical Systems, Technical University of Munich</i> <i>PyTorch, cuRobo, Newton (NVIDIA Warp), CUDA, Python</i> • Built a neural-network-driven pipeline to optimise robot arm designs for confined-space tasks, achieving $\approx 7\times$ speedup over gradient-based IK baselines • Developed collision-aware inverse kinematics and motion planning validation with cuRobo	Munich Apr 2026 – Aug 2026
IPRAKTIKUM WITH SCHWARZIT, TECHNICAL UNIVERSITY OF MUNICH <i>Developer, specialising in object detection models and 3D visualisation on VisionOS</i> <i>YOLOv11, CoreML, SwiftUI (ARKit, RealityKit, Raycasting), Python, Dataset Creation</i> • Built <i>Dishcovery</i>, an Apple Vision Pro app that scans ingredients and delivers personalised recipe recommendations • Integrated computer vision for real-time interaction within spatial environments, using ARKit and the Vision framework • Project page and demo video: aet.cit.tum.de/projects/courses/ipraktikum/24w/schwarzit-it	Munich Oct 2024 – Feb 2025

SOFTWARE ENGINEERING EXPERIENCE

VERIDOS GMBH <i>Working Student Test Automation Engineer</i> <i>TestComplete, Swagger, JavaScript, Azure DevOps</i> • Developed and maintained automated test scripts to ensure software quality, boost reliability and quickly identify bugs	Munich Jul 2025 – Jan 2026
AGILE ROBOTS SE <i>Working Student Front End Software Engineer</i> <i>Vue JS, HTML, SCSS, TypeScript, GraphQL</i> • Developed and maintained web-based robotic solutions; conducted code reviews and implemented tests • Co-developed a graphical component library adopted company-wide for software development	Munich May 2024 – Nov 2024

SELECTED PROJECTS

PELICAN PREDICT AI-driven stock analysis platform built at TUM.ai Makeathon 2025, integrating real-time market data, LLM-generated investment insights and dynamic visualisations	React, MongoDB, Python FastAPI
WEB-BASED QR CODE ATTENDANCE SYSTEM In active use by an organisation of around 80 members; integrated with Google Forms, Sheets and Calendar for collection, reporting and event invitations	Next.js, Supabase, Google Apps Script
CRC32 CHECKSUM ALGORITHM Hand-optimised with SIMD intrinsics, developed alongside a unit-test suite and a benchmarking harness Also built: TripTracker (SwiftUI, published on the App Store) · MORI Money Diary (Flutter, Firebase)	C

SKILLS & PERSONAL

Programming: Python, C, Java, TypeScript, JavaScript, Swift, Dart, SQL
Robotics & ML: PyTorch, ROS 2, NVIDIA Isaac Sim / Isaac Lab, Newton (NVIDIA Warp), cuRobo, CUDA, OpenCV, YOLO, CoreML
Web & Tools: React, Next.js, Vue JS, Three.js, Tailwind CSS, HTML/SCSS, Docker (Compose), Git, GitLab, Azure DevOps
Languages: Indonesian (native), Javanese (native), German (fluent), English (fluent), Chinese (basic)
Personal: KMKI Bayern, Publication & Documentation (2023–2024) · Photography, hiking, travel, cooking